CHATHUKA DILAKSHANA

ASSOCIATE SOFTWARE ENGINEER | FULL STACK DEVELOPER

Rathnapura, Sri Lanka | 0703749261| chathukadilakshana33@gmail.com

GitHub: https://github.com/IT22091352 | LinkedIn: http://www.linkedin.com/in/chathuka-dilakshana-006315284

Portfolio: https://chathukadilakshana.vercel.app/

Professional Summary

Full-Stack Developer with hands-on industry experience in ERP systems and scalable web applications using Laravel, MERN stack, and Next.js. Skilled in RESTful API development, AI integration, database design, debugging, and production deployment, with practical experience in AI/ML projects and a BSc in Information Technology from SLIIT. Strong understanding of software development practices, clean code principles, and building user-focused solutions.

Technical Skills

- **Languages**: Java, PHP, JavaScript, Python, SQL, HTML5, CSS3, TypeScript
- **Frameworks & Libraries**: Laravel, React.js, Node.js, Express.js, Next.js, Tailwind CSS, Bootstrap 5
- **Databases**: MySQL, MongoDB, Supabase
- **Tools & Practices**: Git/GitHub, Postman, Prisma ORM, JWT Authentication, SDLC, Agile/Sprint-based Development, Code Reviews, CI/CD (Vercel), QA & Regression Testing
- **Specialized Domains**: ERP Systems Development, Machine Learning, Deep Learning, UI/UX Design, AI Integration

Work Experience

Apps Technologies (Pvt) Ltd | Software Developer Intern *Jan 2025 - Aug 2025*

- Designed, developed, and deployed 3 mission-critical ERP modules (Inventory Reporting, Requisitions, Finance Workflows) consumed via RESTful internal APIs by 5+ teams, using Laravel, Bootstrap 5, and JavaScript.
- Designed relational database schemas and backend logic, including auto-generated transaction reference codes and account structures, to improve data consistency across finance workflows.
- Debugged and resolved 35+ critical production defects across core logistics modules (Stock Issue, Stock Transfer, POS) within sprint-based development cycles, increasing system uptime by 15%.
- Conducted QA and regression testing to identify edge cases, streamlining deployment workflows for smooth, low-risk production releases.
- Built responsive UIs with dynamic dropdowns, pagination, and data export functions, improving reporting efficiency for 50+ end users.
- Participated in code reviews and cross-functional collaboration with QA engineers and stakeholders, authoring technical documentation and delivering live product demonstrations.

Project Experience

AI-Powered Sinhala Sign Language Platform *Jul 2025 - May 2026*

Python, Flask, React.js, NLP

- Developed a React.js interface and custom NLP engine to tokenize Sinhala text and map vocabulary to standardized semantic IDs.
- Engineered a rule-based algorithm to restructure Sinhala grammar into visual-spatial (TSOV) format, and integrated a Word2Vec model for semantic slang matching.
- Optimized real-time video generation with MoviePy, streaming multimodal 60 FPS animations in under 200 milliseconds.
- Co-authored and published research findings in **IEEE Xplore** (eSmarTA-2026)

Operations Hub – Project Management Platform *Sep 2026*

Next.js 15, Node.js, Express.js, Prisma ORM, MySQL, TypeScript, Tailwind CSS

- Built and launched a full-stack project and task management platform with role-based access control (RBAC) across Admin, Manager, and Member dashboards, **deployed live at project-management-cyphlab.vercel.app**.
- Architected a RESTful API with Prisma ORM and MySQL to manage multi-tier relational schemas, atomic task assignment transactions, and secure JWT-based authentication.

Deep Learning Cancer Data Analysis *Apr 2026*

CNN, TensorFlow, Keras

- Benchmarked four deep learning architectures (including ResNet-DNN and Wide & Deep Network) for breast cancer classification on the Wisconsin Diagnostic Dataset.
- Achieved 98.8% classification accuracy with zero false negatives (1.00 malignant recall), meeting the critical clinical safety target.

ViralFlow AI (SaaS Platform) *Apr 2026*

Next.js, Node.js, MongoDB, Tailwind CSS

- Architected and launched a live SaaS MVP for automated video transcription and subtitle generation, deployed to production at tryviralflow.vercel.app.
- Built a scalable full-stack application using Next.js App Router and Node.js, integrating AI-driven processing pipelines for text generation.

IoT Intrusion Detection System *May 2025*

Python, XGBoost, Machine Learning

- Built a data pipeline to preprocess the 1.5M-record CICIoT2023 dataset using RobustScaler and class-balancing techniques.
- Trained a GPU-accelerated XGBoost model classifying 33 cyber-attack categories, achieving 99.37% accuracy and an 82.00% Macro F1-score on unseen data.

Community Empowerment Hub *Apr 2025*

MERN Stack, OpenAI API

- Built a marketplace and job portal with role-based access control for Sellers, Buyers, and Admins, secured with JWT authentication.
- Integrated OpenAI APIs to deliver AI-powered tools that enhanced platform utility and user interaction.

Ceylon Buddy – Tour Guide Web Application *Apr 2025*

MERN Stack

- Designed and deployed a fully responsive tour guide platform for a client, live at ceylonbuddytour.com, ensuring scalability and security.

Tourist Management System *Dec 2024*

MERN Stack

- Developed a full-stack tourism management application with secure CRUD operations for bookings, travel packages, and authorization protocols.

Education

BSc in Information Technology *(2022 – 2026 | Graduated)*

Specializing in Information Technology — Sri Lanka Institute of Information Technology (SLIIT)

G.C.E. Advanced Level — Maths Stream, St. Aloysius College, Rathnapura *2020*

References

Mr. Buddika Harshanath — Senior Lecturer, Dept. of Information Technology, SLIIT | harshanath.s@sliit.lk

Mrs. Dushanthi Sadeepa Kuruppu — Lecturer (Tenure-Track), Faculty of Computing, SLIIT | dushanthi.k@sliit.lk